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American Carrier Strike Groups: An Electronic Perspective



Fig. 1: A typical US Navy carrier strike group



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An aircraft carrier is a moving airbase that can house approximately hundred combat aircraft. Being a ship, it can also sail to its war theatre quickly. By remaining in international waters, it can launch punitive airstrikes on enemy targets. A combat aircraft can drop two tons of bombs in one go. So, hypothetically, 100 aircraft from a carrier can drop 200 tons of bombs on the enemy's head, at a time.

A century ago, after the invention and eventual militarisation of the aircraft, somebody from the navy thought, "Why do we not operate aircraft from larger ships?" This idea led to the construction of a cantilever on a ship, from which the aircraft took off and landed on.

This resulted in the construction of dedicated aircraft-operating ships, which were called aircraft carriers or, simply, carriers. But these were not taken seriously, initially, and were used only as escorts or scouts to battleships, because during that era, combat aircraft itself was not considered an effective weapon altogether.

This perception changed during World War II as Luftwaffe (German Air Force) taught damaging lessons to allied nations about the combat prowess of aircraft.

From the other side of the globe, Japan delivered a devastating aerial strike at Pearl Harbour using their carrier based aircrafts. In one stroke, they almost entirely decimated the Pacific fleet of the US Navy (USN) but missed to destroy US navy's carriers.

USN, after losing powerful battleships, had to retort to aircraft carriers as primary fighting weapons. It urgently assembled various ships to form taskforces with carriers as centrepieces and other ships to protect the carriers. These taskforces proved their mettle in the subsequent Battle of Midway and numbed the supremacy of Imperial Japanese Navy (IJN) in the Pacific. From that instant, aircraft carriers took the driving seat in naval warfare.

Years after World War II, these taskforces morphed into formidable naval forces called carrier strike groups (CSGs). CSGs pack enormous strike power but, at the